

Push, Scan, Tap: Embracing Commodity Technology for the provision of Accessible Information in Cultural Institutions

Ruth Nagassa
Monash University
Melbourne, Australia
ruth.nagassa@monash.edu

Matthew Butler
Monash University
Melbourne, Australia
matthew.butler@monash.edu

Leona Holloway
Monash University
Melbourne, Australia
leona.holloway@monash.edu

Erica Tandori
Monash University
Melbourne, Australia
erica.tandori@monash.edu

Kim Marriott
Monash University
Melbourne, Australia
kim.marriott@monash.edu



Figure 1: Voice recording push button; tactile QR code; and tactile NFC tag adjacent to wall text in the exhibition space.

ABSTRACT

Cultural institutions often employ commodity technology solutions for the provision of information in exhibitions. While these solutions provide a pragmatic approach, the efficacy, accessibility and user experience for people who are blind or have low vision (BLV) remains understudied. We conducted a case study with an Australian regional gallery to investigate BLV visitors' use of commodity solutions – QR and NFC technology, voice recording push buttons, and touch-triggered electronics – and their efficacy in supporting the provision of accessible information. Through in-situ evaluation, post-visit questionnaire, and semi-structured interviews we present a deeper understanding of the needs and preferences of BLV visitors. Our findings illuminate key considerations to guide the meaningful use of commodity technologies by cultural institutions for accessibility. We hope that this case study will lead to more accessible solutions in cultural spaces, ensuring they more closely meet the needs of the BLV community and encourage more widely-adopted solutions.

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CCS CONCEPTS

• Human-centered computing → Accessibility technologies.

KEYWORDS

blind, low vision, accessibility, art, inclusive design

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1 INTRODUCTION

Cultural institutions are rich with visual artefacts that are often inaccessible by visitors who are blind or have low vision (BLV). Efforts for equitable experience through the provision of braille, large print and audio descriptions are not yet pervasive, and dedicated accessible tours are infrequent [16]. Thus there is little support for spontaneous visitation by BLV people. Further, there is evidence to suggest that much of the BLV community is still disengaged with the creative arts [2, 18].

Accessibility researchers are increasingly turning their attention to solutions for more accessible cultural institutions. This has led to multi-modal technological approaches that have provided new avenues for equitable access to information [17]. Research

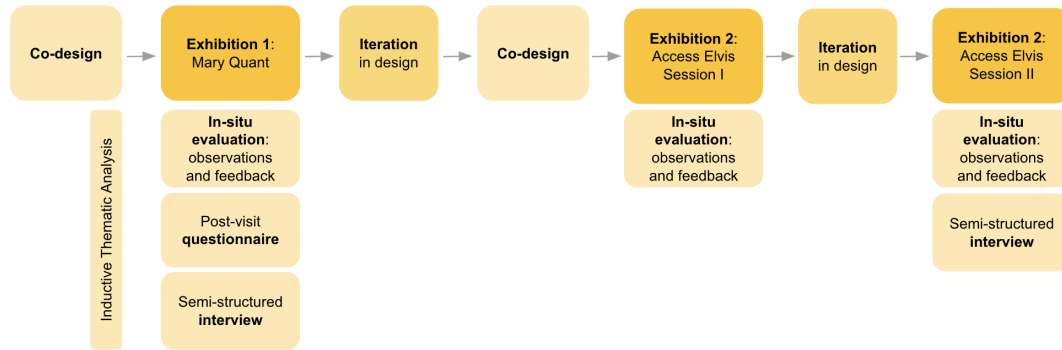


Figure 2: The study procedure involved co-design sessions with BLV colleagues, three sessions across two flagship exhibitions, in which an iterative design process was adopted, followed by thematic analysis of the results.

efforts have utilised smart devices for the provision of accessible information related to artworks [8, 9]. Further research efforts for equitable access have included: direct 3D representation of artefacts [5, 7], interactive 3D models of heritage sites [6, 10], and interactive indoor maps [20].

Cultural institutions across Australia [1, 12, 13, 15] and abroad [14, 19] make use of commodity technology solutions for the provision of information in exhibitions. By ‘commodity technologies,’ we refer to tools that are both widely available and affordable, such as QR and NFC technology. Today, these technologies are often found alongside artworks in exhibitions to provide relevant text or audio content to interested visitors. Such commodity technologies are simple tools, not reliant on advanced technology or infrastructure. Therefore, these technologies are often more affordable and scalable. The prevalent use of such solutions indicates a pragmatic approach by the cultural institutions. **However, these solutions rarely prioritise accessibility and often rely on sight [16]. Although these commodity technologies are widely-adopted, we lack a deeper understanding of how they are used by people who are BLV and their effectiveness in supporting the provision of accessible information in cultural institutions.**

This case study was part of a larger research project [4] that was conducted in collaboration with the Bendigo Art Gallery, an internationally renowned art gallery in regional Australia. Here, we delve further into the use of commodity technology – QR and NFC technology, voice recording push buttons, and touch-triggered electronics – to understand their efficacy and use by BLV people in real-world situations.

This case study makes two primary contributions: (1) it presents an in-the-wild study across two flagship gallery exhibitions to better understand the use of QR and NFC technology, voice recording push buttons, and touch-triggered electronics and an analysis of the benefits and compromises for BLV accessibility; (2) provide guidance for the use of these technologies for accessibility by cultural institutions and practitioners in the community.

2 METHODOLOGY

2.1 Method Overview

We collaborated with Bendigo Art Gallery to produce a variety of accessible solutions for BLV visitors for their 2021 and 2022 flagship

exhibitions: *Mary Quant: Fashion Revolutionary* and *Elvis: Direct from Graceland*. In three sessions across these two exhibitions, we explored the commodity solutions in two distinct ways: first as a means of accessing artefacts through interactive 3D printed models (I3DPMs); and second, as a means of accessing wall texts throughout the exhibition space. The two flagship exhibitions were held approximately 12 months apart, facilitating an iterative design process in which designs were amended based on observations and feedback from each session, as explained in results section 3.1. As outlined in table. 1, four BLV visitors attended Exhibition One (Mary Quant), with three participating in the questionnaire and semi-structured interview. Twenty BLV visitors attended Access Elvis Session One of Exhibition Two, and sixteen BLV visitors attended Access Elvis Session Two. A total of eight participants from Access Elvis Sessions One and Two took part in semi-structured interviews.

2.1.1 Co-design of commodity technology solutions. The gallery provided briefs prior to the exhibitions that detailed the items and information to be exhibited. We conducted virtual and in-person meetings with a colleague who is blind for the ideation of ways in which commodity solutions could be made more accessible. This involved incorporating tactile markers at information access points, using key items from the exhibitions, such as a daisy and record label. These tactile markers were integral in BLV users locating the position of the audio access points and allowed for a cohesive integration with the exhibitions. The research team also consisted of a legally blind artist, an experienced tactile graphics transcriber and researchers experienced with the use of emerging technologies for accessibility.

2.2 Interactive Materials

Four types of easily available commodity technologies were chosen for use: Quick Response (QR) codes; Near-Field Communication (NFC); voice recording push buttons; and touch-triggered electronics. The technologies were applied to I3DPMs, a large diorama, as well as to accompany standard wall text.

2.2.1 Touch-Triggered Electronics. Touch-triggered electronics were used on a 3D printed Mary Quant fashion model (fig.3). A single touch input and audio feedback output was controlled using a Bare

	Mary Quant	Access Elvis Session I	Access Elvis Session II
Visitors	4	20	16
Participants	3		8
Commodity Technology	Touch-triggered electronics; QR & NFC technology	QR & NFC technology	QR & NFC technology; voice recording push buttons

Table 1: Number of visitors, participants and types of commodity technology used across three sessions in the two exhibitions.



Figure 3: Left: interactive 3D printed fashion models of mannequin poses from the Mary Quant exhibition. Each model had one of either a QR code, NFC tag or touch-triggered electronics. Right: 3D printed model of Elvis' first home accompanied by voice recording push buttons and a QR code.

Conductive Touch Board ¹ and an external speaker. The electronics were compartmentalised and a conductive disk served as the touch point in the centre of a tactile daisy.

2.2.2 QR & NFC Technology. Both QR and NFC technology were explored for I3DPMs and wall texts. The QR and NFC technology were linked to accessible multimedia content and collated exhibition information provided by the gallery. The linked web page included structured text that could be read aloud with a screen reader, high-contrast colours and music from the decade to provide an aesthetic impression. QR codes and NFC tags (NTAG213) were placed beneath a tactile marker.

2.2.3 Voice Recording Push Buttons. Voice recording push buttons were introduced as a simpler solution in Access Elvis Session Two, to accompany the diorama (fig. 3: right) and wall texts (fig.1). It could be triggered with a push of the large coloured button, with sound emanating from the button itself.

3 RESULTS

3.1 Iteration in Design

Interestingly, during exhibition two, we observed that some participants would scan the QR code, then refrain from touching the I3DPMs and solely listen to the audio on their personal device. One reason came down to preference:

“I found that I actually didn’t need to touch the diorama [I3DPMs], you could hear the story of it.” [P4]

Following this observation, we introduced voice recording push buttons and updated the audio content to encourage tactual enquiry of the I3DPMs, which led to a richer experience in Access Elvis Session Two:

¹<https://www.bareconductive.com/collections/all/products/touch-board>

“The addition of information buttons was very good and helped to explain and guide the activity.” [P6]

Further, the updated audio content was valuable in supporting understanding, particularly of the I3DPMs. Indeed, as explained by a visitor, it not only supported their mental model but was able to convey more than tactile information alone:

“I liked the buttons that talked, they tell a story, I could imagine it in my head... that tells me more than feeling something.” [P4]

3.2 Thematic Analysis

As illustrated in fig.2, data was captured from all three sessions. This included the recordings of interviews, questionnaire responses, as well as in-situ observations and feedback. We analysed a specific subset of the data reported in [4] focusing particularly on aspects related to the commodity technologies used. For example, this involved interview questions about participant attitudes towards the labelling technologies and their usefulness in their engagement with the exhibition. Additionally, we considered relevant findings from general feedback on the exhibition. Responses and observational data were transcribed and an inductive thematic analysis approach was adopted. Specifically, two researchers independently coded the transcripts. Once all transcripts were coded, together the researchers then reviewed, cross-checked and revised these codes. The codes were collectively reviewed, sorted, and collated by all researchers to generate a set of themes. The final themes are described.

3.2.1 Theme 1: Enhancing accessibility with immediate solutions, complementing independent and guided visitor experiences. The overall feedback was overwhelmingly positive and the commodity solutions provided much-needed accessible information. It was clear that the use of simple technologies, immediate accessibility and the availability of alternative formats enhanced the accessibility of information. As explained by a support worker of a BLV visitor, the commodity solutions successfully provided audio labels that were useful and supported independent access to gallery content:

“That was absolutely necessary. Very useful. She could listen and I’m not trying to read everything to her.” [P8]

Specifically, as explained by a visitor with low vision, the use of high-contrast colours was a valuable design choice.

“These buttons you reached up to, I can see yellow against a background really well.” [P3]

Further, the tactile markers were imperative to convey the audio label access points, particularly on I3DPMs:

“So her daisy [tactile marker] on the front, you can touch it and it played here.” [P11]

While sighted guidance was reported very positively and considered sufficient without additional support, the benefit of commodity solutions was still acknowledged:

“Audio labels not required as had assistance, but would use in future.” [P1]

Others who attended with a sighted guide benefited from the support when using the commodity solutions. Researchers were located throughout the exhibition space to inform visitors of the location, use and purpose of the commodity solutions.

Indeed, locating the audio label access points accompanying the wall texts presented some challenges to visitors. Some level of guidance was needed to navigate the space, locate its position on the wall and retrieve the audio label. For this reason, internal orientation systems such as NaviLens®² were suggested to guide visitors to the audio access points:

“I also feel that I missed a lot of the information buttons and QR codes because I can’t see where they are. If an internal system was developed where it was possible for a blind person to move around alone, the position of these buttons and tags could be built into the orientation system.” [P7]

As were tactile ground surface indicators:

“I also wonder whether another option would be to put some sort of tactile marking on the floor so that blind people could follow the line of the exhibition independently.” [P8]

Further, sighted guidance was also needed to assist in locating and capturing QR and NFC technology. A specific challenge observed with the QR codes was difficulty using the camera to capture the code:

“I was not aware of how to use it, such as if I needed to open my camera app or not and needed advice from others on this, and where to place the camera in relation to the code.” [P10]

This may be rectified by apps such as VIP Code Reader³, a QR scanner designed for use by people who are BLV that indicates the position of QR codes on vertical and horizontal surfaces.

3.2.2 Theme 2: Prioritising simplicity and universal accessibility. Technological solutions that are intuitive, provide immediate feedback, prioritise universal access and rely on minimal devices were preferred. The voice recording push buttons were most preferred by many due to the ease of use and the immediacy of the response:

“The push buttons were easy, just because, when you pushed it, you got an immediate response. This was certainly the simplest and most effective method of obtaining information.” [P7]

Touch-triggered electronics were also preferred for their simplicity and universal accessibility:

“I prefer the speaker model for its simplicity and availability to people of all abilities.” [P11]

One reason visitors preferred both voice recording push buttons and touch-triggered electronics was that, unlike the other technologies, it did not require the use of a smartphone:

“I prefer the push button model the most because it is easy to use and does not require juggling a phone which can be tricky when you are holding a cane or Guide Dog lead.” [P10]

Further, these technologies freed both hands to touch read the I3DPMs:

“You haven’t got to try and handle your phone and the model, you can actually just listen to the description while you’re actually touching the model.” [P11]

3.2.3 Theme 3: Offering flexibility and customisation. Considering the environmental setting is vital. For instance, the loud environment within the gallery was a major challenge noted by many of the visitors:

“It was very noisy and I couldn’t hear my phone.” [P7]

Subsequently, many participants mentioned being overwhelmed by the noise and crowds and felt a time pressure. Unsurprisingly, a common thread was the desire to access the information in a more comfortable setting either before or after visiting the gallery, which the QR and NFC technology solutions facilitated:

“The thing I liked is that you have it on your phone and then you can read it at home.” [P6]

Providing a combination of remote and in-person access to information offers the flexibility for visitors to make use of the advantages of both [11]. This is particularly true for BLV visitors during opening hours where there are more environmental and navigational challenges as opposed to dedicated accessible sessions [3].

“I love the idea, if you could save those for when you got home and then revisit.” [P11]

The customisation of solutions was also important. QR and NFC access to digital content did have notable preferred features including the ability to customise speech playback speed and volume:

“The options for the phone could also be beneficial as you can adjust the screen access parameters such as volume and rate of speech to your own preferences.” [P10]

Individual preferences for visual art access methods varies widely among the BLV community [11]. Providing a range of solutions or customisable features may address this. Each commodity solution offered unique advantages and were particularly useful in specific contexts. For instance, digital content provided by QR and NFC technology allowed people to seek out further information and revisit this information at a time of their convenience. Although this work did not seek to understand the benefits of implementing multiple technologies simultaneously, we believe it was beneficial to visitor experience as they had the freedom to pick the technology that they were most comfortable with.

3.2.4 Theme 4: Presumptions of technological expertise and ownership of smart devices. Presumptions of visitor familiarity with technology, technological expertise and ownership of a smart device can present barriers to access. It is crucial to critically consider

²<https://www.navilens.com/en/>

³<https://apps.apple.com/us/app/vip-code-reader-blind-scanner/id1273427754>

any prerequisites for technology use in order to prioritise inclusivity.

Familiarity with the technology was a major factor in visitor experience. As explained by a visitor, the increased presence of QR code check-ins during the COVID-19 pandemic meant that many people had some exposure to the technology:

“I have spent the last few months learning how to use QR codes for entering premises and therefore didn’t find it scary or difficult. It was also great that the information was immediately available on my phone.” [P11]

NFC tags were the least familiar of the technologies and required visitors to own an NFC-compatible device, which was not always the case:

“I’m not very familiar with the technology, so was not good at obtaining the information.” [P7]

Indeed, we observed that due to a lack of familiarity, accessing information through NFC tags presented the most challenges for BLV visitors. Further, a small number of participants mentioned their technological expertise as a barrier:

“I’m not a fantastic user of Safari, so it took a couple of seconds to find the information within the web page.” [P11]

Practice sessions were suggested as a method of introducing visitors to the technology prior to exploring an exhibition:

“This [QR code] is not a system I’m very familiar with and therefore I was not as effective or efficient as I would have liked. I think this sort of information giving is something people should have an opportunity to practice before they go to an environment.” [P7]

Ideally, the time that visitors spend in cultural institutions should be centred around engaging with cultural works and not learning new technology in order to do so. Providing information about the accessibility solutions online and onsite at entry points would provide visitors with the opportunity to first become familiar with the technology. This ensures that any necessary learning can be done at a time of convenience.

Drawing from the aforementioned themes, we determined that the commodity technologies were most suitable for the following contexts:

- **QR and NFC technology:** These were best suited to support access to wall texts and to support the provision of information either prior to or after visiting a cultural institution. Given their widespread familiarity and the ability of most devices to natively scan QR codes, we recommend the use of QR codes over NFC technology.
- **Voice recording push buttons and touch-triggered electronics:** These technologies were found to be highly suitable for I3DPMs as they allowed users to free both hands for touch reading. While these solutions were valuable for wall texts, it’s worth noting that they may offer less flexibility and customisation compared to other options. Considering the real-world environmental challenges, we recommend using them in conjunction with headphones.

4 IMPROVING THE ACCESSIBILITY OF COMMODITY TECHNOLOGY IN CULTURAL INSTITUTIONS

This case study is of importance as it provides valuable insight into the accessibility of commodity technology commonly used for the provision of information in cultural institutions. While commodity solutions support the provision of information, there’s room to enhance their accessibility by considering the varied needs and preferences of the BLV community. We offer the following considerations as guidance to improve the accessibility of commodity solutions:

- **Diversity in Solutions:** It’s crucial to recognise the variety of needs and preferences among the BLV community. Hence, providing multiple commodity technological options will allow visitors to choose the tool that works best for them.
- **Well-integrated Solutions:** Any commodity solution should complement, not distract from the purpose. For instance, the use of smart devices to support the use of I3DPMs should be carefully considered, particularly when tactual inquiry is a priority.
- **Navigation & Capturing of QR and NFC Technology:** Clear navigation to any commodity solutions within a space is crucial. When applicable, guidance should also be provided on how to capture QR and NFC technology with any smart device.
- **Support & Guidance:** While commodity solutions enhance independent exploration, their introduction should complement, not replace, the role of sighted guides or gallery experts. Moreover, having staff or researchers available to explain and assist with the use of these commodity solutions can greatly improve the visitor experience.
- **Intuitiveness & Simplicity:** Solutions that are easy to use, provide immediate feedback, and require minimal devices are preferred. Hence, technologies like voice recording push buttons and touch-triggered electronics that give instant feedback can significantly enhance the user experience.
- **Customisation:** Solutions should support customisation of features by users according to their needs and preferences, particularly in the case of digital content.
- **Pre-visit and Post-visit Access:** Solutions that allow visitors to access content at their convenience, before or after visiting the cultural institution can enhance experience.
- **Technology Familiarity:** Solutions should be universally accessible without assuming a certain level of technological proficiency. Given that visitors might have varying degrees of familiarity with technologies like QR and NFC, there should be clear instructions, support and opportunity for familiarisation with the technology before entering the exhibition.

With the widespread adoption of commodity technology in cultural institutions today, there’s a clear opportunity to make these solutions more accessible. By taking the above mentioned considerations into account, cultural institutions can make practical adjustments to the commodity technology already in use. This, in

turn, will enable them to offer more immediate accessibility solutions and hopefully encourage spontaneous visitation to cultural institutions by BLV people.

5 LIMITATIONS

While the qualitative feedback gathered in this case study is invaluable, it would have been beneficial to have quantitative data to objectively measure the usage and effectiveness of the commodity solutions. A larger follow-up study will allow for further exploration of the identified themes.

6 CONCLUSION

In this case study, we investigated the use of QR and NFC technology, voice recording push buttons, and touch-triggered electronics, for the provision of accessible information in two major exhibitions. Our findings illuminate that such commodity technologies are indeed useful and present an opportunity for cultural institutions to implement simple yet effective accessibility solutions. This work presents key considerations as guidance to improve the accessibility of commodity solutions. Due to their wide availability and low-cost, we recommend that commodity technologies are further embraced by cultural institutions and that a diversity of solutions are adopted to prioritise choice by BLV visitors. We hope that this case study will improve the accessibility of commodity technology commonly found in cultural institutions, lead to research that is more relevant to the needs and preferences of BLV people and encourage more widely-adopted solutions.

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